

APPENDIX 4:

INTERNSHIP REPORT TEMPLATE

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EXECUTIVE SUMMARY (200 words)

This report summarizes my 16-week internship experience at Phenikaa-X, a leading Vietnamese high-tech enterprise specializing in autonomous systems, robotics, and smart mobility solutions. Throughout the internship, I gained both theoretical knowledge and hands-on experience in UAV and mobile robot systems, focusing on PX4 flight control, ROS2 integration, and computer-vision-based navigation.

The primary project identified during the internship addressed a key challenge in robotics—the gap between simulation and real-world performance. Systems that performed flawlessly in simulation often showed instability and reduced accuracy in actual operation due to environmental noise, lighting variation, and sensor calibration errors. To solve this, a closed-loop testing and calibration framework was proposed, combining improved simulation realism, real-world data feedback, and adaptive control to enhance performance consistency across environments.

Through this internship, I also strengthened essential engineering skills, including embedded programming, ROS2 development, and system integration, while recognizing areas for further improvement in algorithm design and manufacturing practice. Additionally, I learned the importance of ethical and professional responsibilities, such as safety, data privacy, and regulatory compliance.

Overall, the internship provided valuable technical experience, professional growth, and a strong foundation for future research and development in autonomous robotics.

INTRODUCTION

Introduction of the host company (500 words): Phenikaa-X Joint Stock Company, headquartered in Hanoi, Vietnam, is a technology company and a subsidiary of the Phenikaa Group, which has diverse interests in industrial manufacturing, technology, scientific research, education, healthcare, and trading. Established in 2020, Phenikaa-X was founded to commercialize research and development (R&D) outcomes from Phenikaa Group and Phenikaa University, with a focus on autonomous systems and smart mobility solutions. The company's mission is to develop high-tech products-such as autonomous vehicles, drones, intelligent robots, and automation equipment-that meet international standards and contribute to the global competitiveness of Vietnamese technology.

Phenikaa-X has grown rapidly since its inception, achieving several key milestones. In 2021, it unveiled Vietnam's first Level 4 "Made-in-Vietnam" autonomous vehicle, marking a significant step in the country's autonomous technology development. In 2024, the company implemented its AMR (Automated Mobile Robot) Pallet Mover solution at the Samsung Electronics Thai Nguyen Factory, making it the first Samsung factory globally to deploy that technology. The same year, Phenikaa-X participated in the Vietnam Manufacturing Expo (VME) 2024, showcasing intelligent automation solutions integrated with IoT and international safety standards, and was recognized at the Top Industry 4.0 Vietnam awards for innovation, smart digital products, and leadership. Strategic partnerships, such as cooperation with Vietnam Post Logistics, demonstrate the company's commitment to integrating autonomous technology into industrial and service applications.

The company's main business areas include autonomous mobility and smart vehicles, developing autonomous minibuses and intelligent transport solutions; robotics and automation for manufacturing and logistics, including AGVs, AMRs, and service robots; drone and UAV systems for monitoring and smart city applications; and R&D, integration, and custom solutions, providing AI, IoT, LiDAR, and cloud-based services. Phenikaa-X also focuses on exporting Vietnamese high-tech products to international markets.

Phenikaa-X's organizational structure reflects its technology-driven focus, with departments likely including: R&D (autonomous vehicles, robotics, mapping, sensors); Robotics & Automation (AMR/AGV development and integration); Mobility & Smart Vehicle Division; Drone & UAV Division; Commercial/Sales & Business Development; Integration & Solutions; Operations & Maintenance; and Administration/HR/Finance. This structure supports a seamless transition from research to production, deployment, and client support.

During my internship at Phenikaa-X, I was placed in a high-tech environment focused on autonomous systems, robotics, and smart manufacturing solutions. The company's commitment to R&D, innovation, and commercialization provides an ideal context for applying theoretical knowledge, such as motion planning and obstacle avoidance in UAVs and mobile robots, into practical industry projects. Working at Phenikaa-X also offers insight into how advanced technology companies organize R&D, production, integration, and client services while maintaining international standards.

Overall, Phenikaa-X represents a pioneering Vietnamese high-tech enterprise that combines research excellence, innovation, and commercial application, providing a rich and dynamic learning environment for my summer internship.

Overview of the internship (300 words): My internship at PNKX spanned a total of 16 weeks, providing a comprehensive exposure to quadrotor systems, both hardware and software. The program began with an orientation on the first day, where I familiarized myself with the internship rules and essential equipment. In the first two weeks, I gained foundational knowledge about quadrotors, learning their hardware components, software architecture, and basic control using PX4. Weeks three and four were dedicated to becoming comfortable with Linux, installing Ubuntu 22 and ROS2, and practicing simulations using Turtlesim.

Week five focused on PX4 installation and Gazebo simulations, alongside studying MAVLink communication protocols and MAVROS integration. During weeks six and seven, I engaged in OFFBOARD mode control outdoors, performing takeoff, circular, and figure-8 flights both in simulation and real-world tests. Week eight introduced PX4's no-GPS mode and alternative positioning methods, while week nine involved installing and utilizing Indoor Positioning Systems (IPS) and AprilTags for indoor localization.

In weeks ten and eleven, I applied OFFBOARD control indoors, maintaining position and executing circular and figure-8 maneuvers. Week twelve focused on precision landing indoors using AprilTags, and week thirteen covered obstacle avoidance and collision prevention using 2D Lidar in simulation. Weeks fourteen and fifteen emphasized path planning with OFFBOARD control implemented on a Raspberry Pi.

The final week was devoted to report finalization and preparation for the end-of-term seminar, culminating in the presentation of outcomes, submission of a Git repository, and user manuals on the final day.

Throughout the internship, I developed both technical and practical skills, including ROS2 programming, PX4 configuration, simulation and real-world flight control, and sensor-based navigation. I also gained experience in troubleshooting, integrating hardware and software components, and collaborative problem-solving. This structured, hands-on program enhanced my understanding of autonomous UAV systems and prepared me to apply these competencies in real-world robotics applications.

Key Date	Key Learning Activities
Day 1	Introduction to the internship program, rules, and equipment orientation
Week 1 & 2	Learn about quadrotors, hardware & software systems at PNKX; basic control using PX4
Week 3 & 4	Get familiar with Linux, install Ubuntu 22 & ROS2, practice with Turtlesim
Week 5	Install PX4, simulate flying with Gazebo, study MAVLink & mavros integration
Week 6&7	OFFBOARD mode control (outdoor): takeoff, circular & figure-8 flight (sim & real test)
Week 8	Study PX4's no-GPS mode and alternative positioning methods
Week 9	Install & use Indoor Positioning System (IPS) or AprilTag for indoor localization
Week 10&11	OFFBOARD mode control (indoor): hold position, circular & figure-8 flight (sim & real)
Week 12	Precision Landing control indoor using AprilTags
Week 13	Obstacle avoidance and collision prevention using 2D Lidar (Sim)
Week 14&15	Path planning problem using OFFBOARD control with Raspberry Pi
Week 16	Finalize report, prepare for end-of-term seminar
Final Day	Present outcomes, submit Git repo, and user manuals

PROBLEM/PROJECT STATEMENT (500 words)

During the internship, one of the most significant and recurring challenges identified was the discrepancy between simulation and real-world application in robotic systems, particularly in autonomous navigation and perception tasks. This issue was observed while working on mobile robots and UAVs that rely heavily on computer vision, sensor fusion, and motion planning algorithms.

In simulation environments, conditions are typically idealized — lighting is uniform, sensor readings are noise-free, and visual data is clean and consistent. Under these circumstances, algorithms such as object detection, pathfinding, and obstacle avoidance tend to perform flawlessly. However, when the same systems are deployed in real-world environments, their performance often drops drastically. This discrepancy stems from the inherent gap between simulated parameters and real-world variability.

For example, the computer vision models used for localization and navigation in the simulation were trained and tested under ideal lighting and camera configurations. In contrast, the real-world camera often suffered from issues such as lens scratches, misalignment, glare, or

low-light conditions. These imperfections resulted in inconsistent image data, reducing detection accuracy and reliability. Similarly, sensors like LiDAR and IMU, which operate with near-perfect precision in simulation, were affected by environmental noise, vibrations, and calibration errors in actual experiments.

In addition to perception errors, motion control also exhibited noticeable discrepancies. In simulation, the robot's dynamics were perfectly modeled, resulting in stable trajectories and accurate landings. However, in real-world tests, drifting, unstable hover, and fluctuating altitude frequently occurred due to imperfect thrust calibration, air turbulence, and motor response delay. Even small differences in propeller balance or payload weight caused cumulative deviations in position control, leading to unstable motion during precision landing and autonomous navigation.

The core issue is not only the difference between simulation and reality but also the lack of robust transferability between them. Systems are often optimized for simulation because it is faster, safer, and cheaper to test. However, without proper calibration, sensor modeling, and environment adaptation, this optimization creates a false sense of reliability that collapses in real conditions. This gap poses significant challenges in robotics development, where the ultimate goal is to achieve consistent real-world performance. Addressing this problem is crucial for improving the reliability and robustness of autonomous systems. It demands not only better simulation fidelity but also smarter adaptation methods that allow systems to handle imperfect and unpredictable environments.

PROPOSED SOLUTION (500 words)

To solve the problem of the big difference between simulation and real-life performance, the proposed solution is to create a closed-loop testing and calibration system that connects simulation results with real-world data. Instead of depending only on simulation, this method focuses on collecting feedback from real tests and using it to improve both control and vision performance. The goal is to make the robot smarter and more adaptable when moving from a perfect simulation to real-world conditions that are often unpredictable.

The first step is to make the simulation more realistic. Many simulations use ideal conditions with perfect lighting and no sensor noise. To fix this, the new simulation setup will include common problems such as uneven lighting, random sensor noise, wind disturbance, and small motor delays. These changes will make the simulation behave more like the real world. By doing so, computer vision and motion algorithms like object tracking, localization, and path planning can be tested and improved under more realistic conditions. This step helps ensure that the robot or drone can handle imperfect situations before being tested in real life.

Next, a data-driven calibration process will be used. During real-world tests, data will be collected from onboard sensors such as the IMU, barometer, and camera. This information helps find common errors like drifting, unstable height, and motor delay. Using this data, the system

can automatically tune control parameters such as PID gains, thrust levels, and altitude corrections in PX4. Repeating this process many times helps the simulation and real-world performance become more closely matched. Over time, the robot's control and response become smoother and more accurate.

For the vision system, a similar improvement will be applied. Instead of using only clear and ideal images from simulation, the vision model will be trained with a mix of real and simulated images. These images will include different lighting conditions, angles, and small camera issues such as blur or glare. This method, called domain adaptation, helps the camera system recognize objects and positions correctly even when the lighting or background changes. As a result, the system will perform more reliably in different real-world environments.

In terms of motion control, an adaptive control layer will be added. This means that the control system can automatically adjust itself when it detects problems such as drifting, unstable hovering, or shaking. For example, if the drone starts losing height or drifting sideways, the controller can increase thrust or correct direction instantly. This helps maintain stable flight and accurate landing without needing constant manual adjustment.

Finally, a step-by-step testing process will be followed. Each improvement will first be tested in simulation, then in small controlled experiments, and finally in full real-world trials. Key results such as accuracy, stability, and smoothness of motion will be recorded after every test.

By combining better simulation, automatic calibration, and adaptive control, this solution helps close the gap between simulation and real life. It will make the robot or UAV more stable, reliable, and ready for practical use in complex environments.

REQUIRED SKILLS (200 words)

During my internship, I realized several technical skills I lacked, which affected my efficiency and highlighted areas for improvement in my fourth year. One major gap was programming. While I had basic coding knowledge, I struggled with algorithm design, debugging, and writing optimized code for practical applications. This limited my ability to implement solutions quickly and efficiently, showing that further study in advanced programming and problem-solving is essential.

Another weakness was practical manufacturing experience. Although I understood theoretical concepts, I had little exposure to real production processes, material behavior, and assembly techniques. This made it difficult to anticipate practical challenges in design and workflow. Gaining hands-on experience through workshops, lab sessions, or industry projects in the fourth year will help bridge this gap.

I also lacked embedded programming skills. Tasks involving microcontrollers, sensor integration, and firmware development were challenging due to my limited exposure. Studying embedded systems, practicing with platforms like Raspberry Pi, and working on real hardware projects will strengthen this area.

Finally, I noticed a need to improve my engineering intuition, the ability to foresee problems and make proactive decisions. Overall, focusing on programming, manufacturing, algorithms, embedded systems, and practical problem-solving in my fourth year will help me become a more competent and well-rounded engineer.

ETHICAL AND PROFESSIONAL RESPONSIBILITIES (500 words)

During my internship, I encountered several situations that highlighted the importance of ethical and professional responsibilities in engineering practice. One significant example involved the development and testing of technical solutions that could directly impact end users and society. For instance, while working on automation and embedded systems projects, I realized that even small errors in design or programming could lead to safety risks or unintended consequences. This made me aware of the broader societal implications of engineering work, such as potential hazards to users, environmental effects, or disruptions to daily operations if systems failed. Recognizing these risks encouraged me to adopt a more cautious and responsible approach, including thorough testing, validation, and documentation of all solutions to ensure they were reliable, safe, and socially responsible.

Another ethical consideration I observed was related to data privacy and security. Some tasks involved handling sensitive information collected from devices and systems. I learned that protecting this data, ensuring confidentiality, and preventing unauthorized access were not only technical requirements but also ethical obligations to safeguard the rights of users. Maintaining integrity in data handling and reporting results accurately reinforced the professional responsibility engineers have toward society, emphasizing honesty, accountability, and transparency in every task.

During the internship, I also became familiar with the standards, regulations, and laws that govern engineering practice in my field. For example, technical solutions had to comply with national safety regulations, industrial standards for electrical and mechanical systems, and environmental guidelines to minimize negative impacts. Adhering to these standards ensured that work met legal requirements, promoted public safety, and upheld professional credibility. In addition, I learned about intellectual property considerations, including respecting proprietary designs and software, which is a key aspect of professional ethics in engineering. Understanding and applying these regulations taught me that ethical engineering practice involves both compliance with formal laws and the broader responsibility to act in the best interest of society.

Overall, my internship experience reinforced that ethical and professional responsibilities are integral to engineering practice. Engineers must not only design effective technical solutions but

also anticipate and address the societal and ethical implications of their work. By recognizing potential risks, respecting safety and privacy standards, complying with regulations, and upholding honesty and accountability, I developed a deeper appreciation for the professional responsibilities inherent in engineering. These lessons will guide my approach in future projects, ensuring that my work benefits society while adhering to ethical principles and legal standards.

() NOTE: Tables or Figures are not included in the word count.*